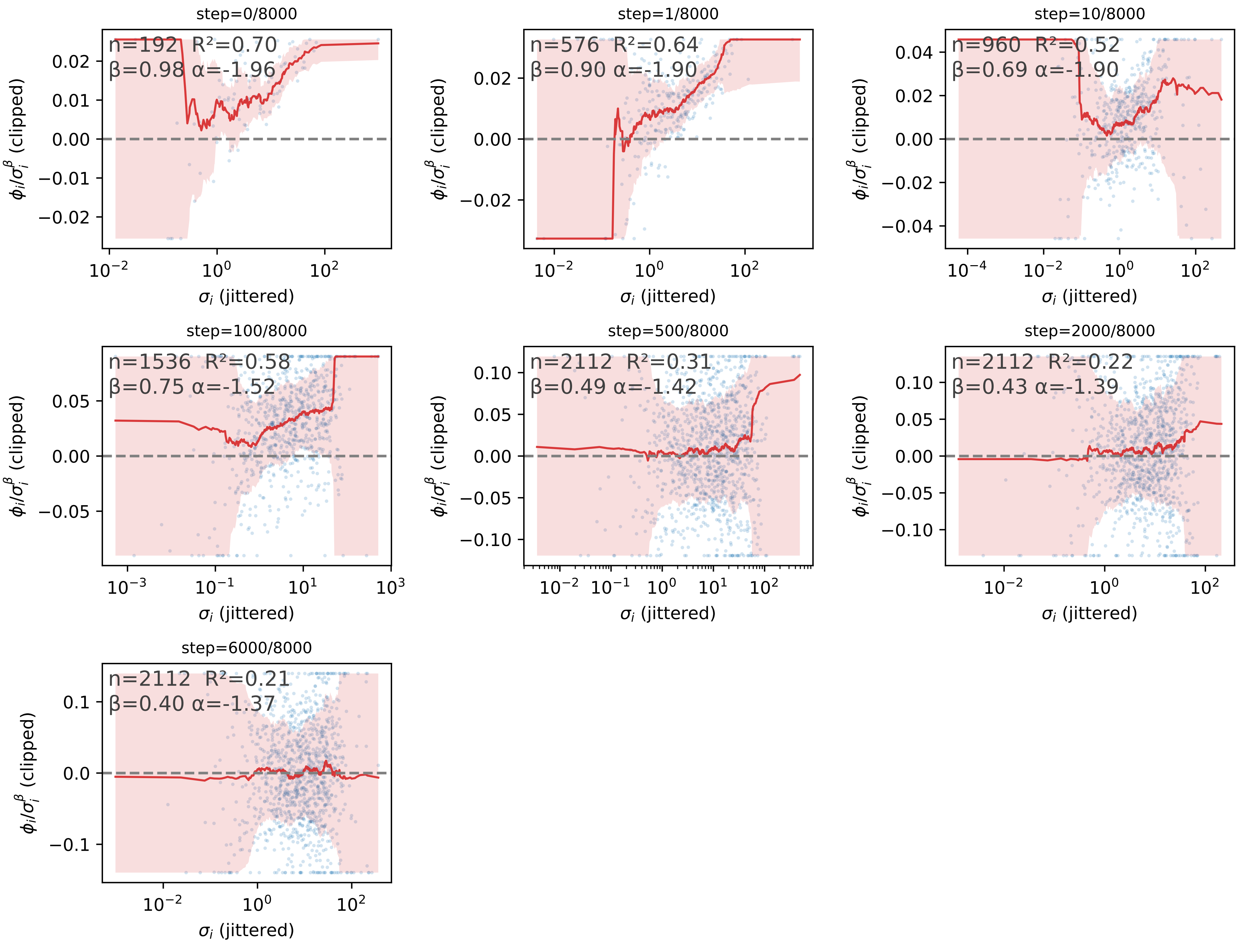
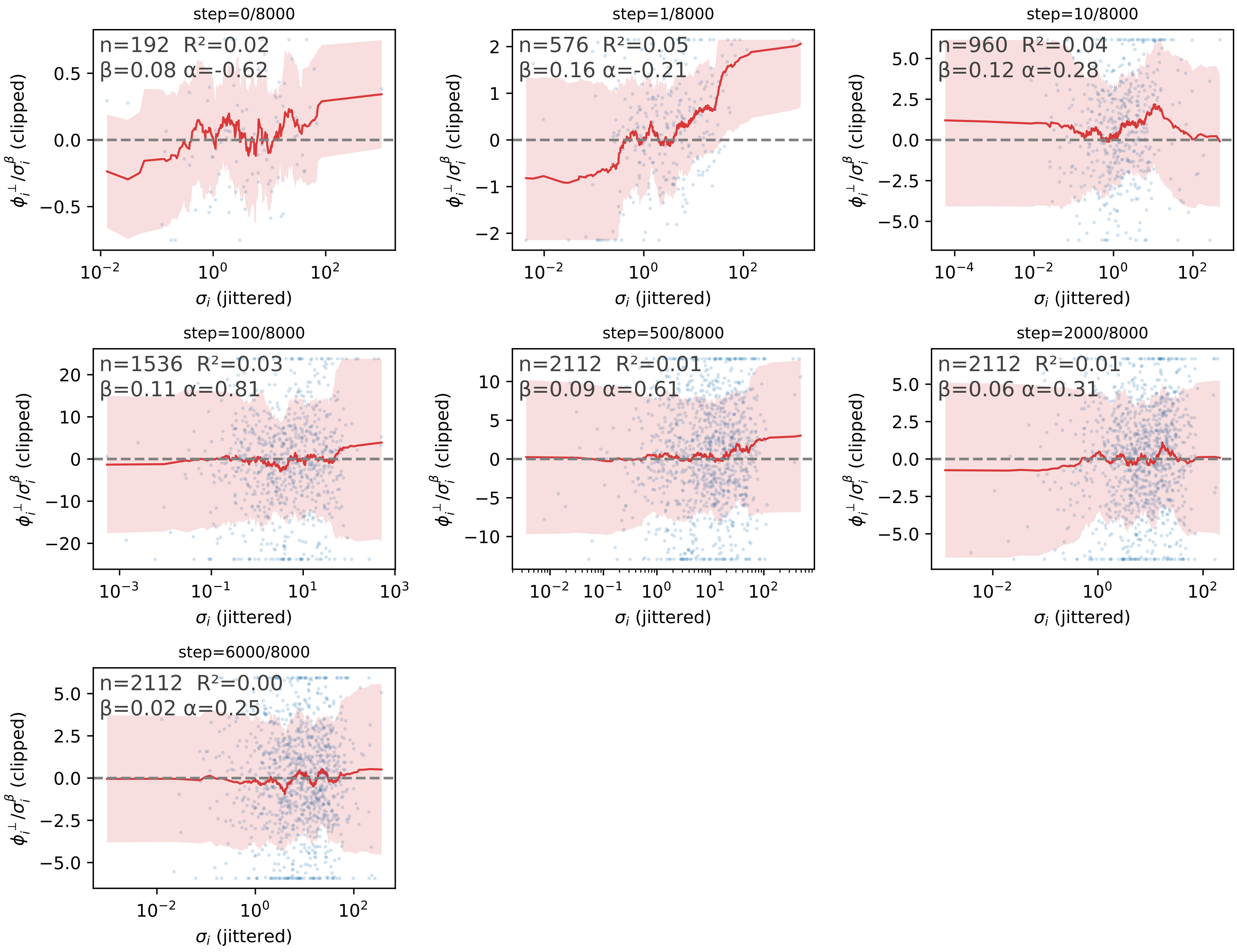
Attention value projection: ϕ_i/σ_i^β vs σ_i (β fit per group from $\log|\phi_i| \sim \beta \log \sigma_i$, one batch-mean point per mode per step in the group)



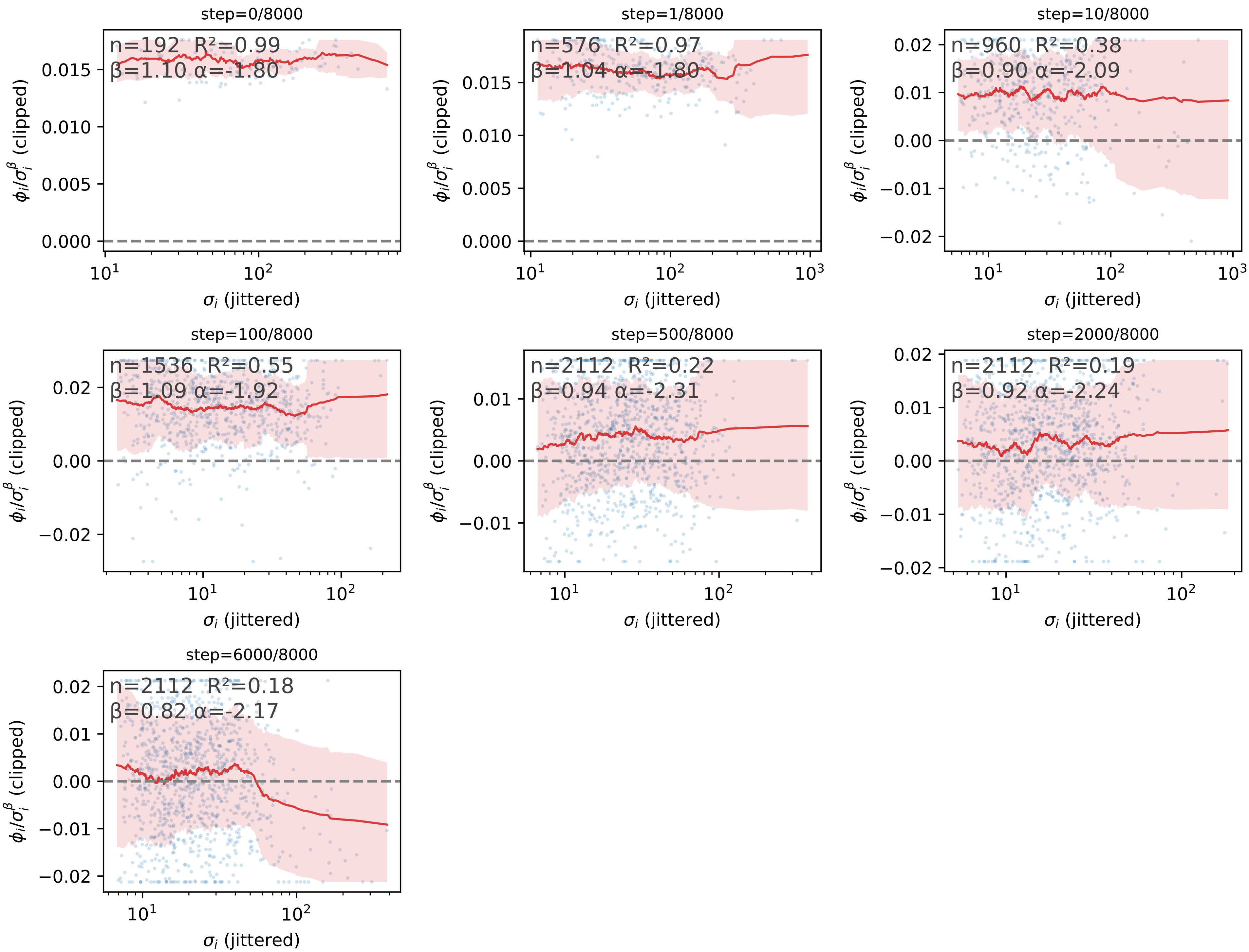
n: number of points.
R^2, β, α: least-squares fit log10(y) = β·log10(x) + α (β the fitted power-law exponent, α the intercept).

Attention value projection: $\phi_i^\perp/\sigma_i^\beta$ vs σ_i (β fit per group from $\log|\phi_i^\perp| \sim \beta \log \sigma_i$, one batch-mean point per mode per step in the group)



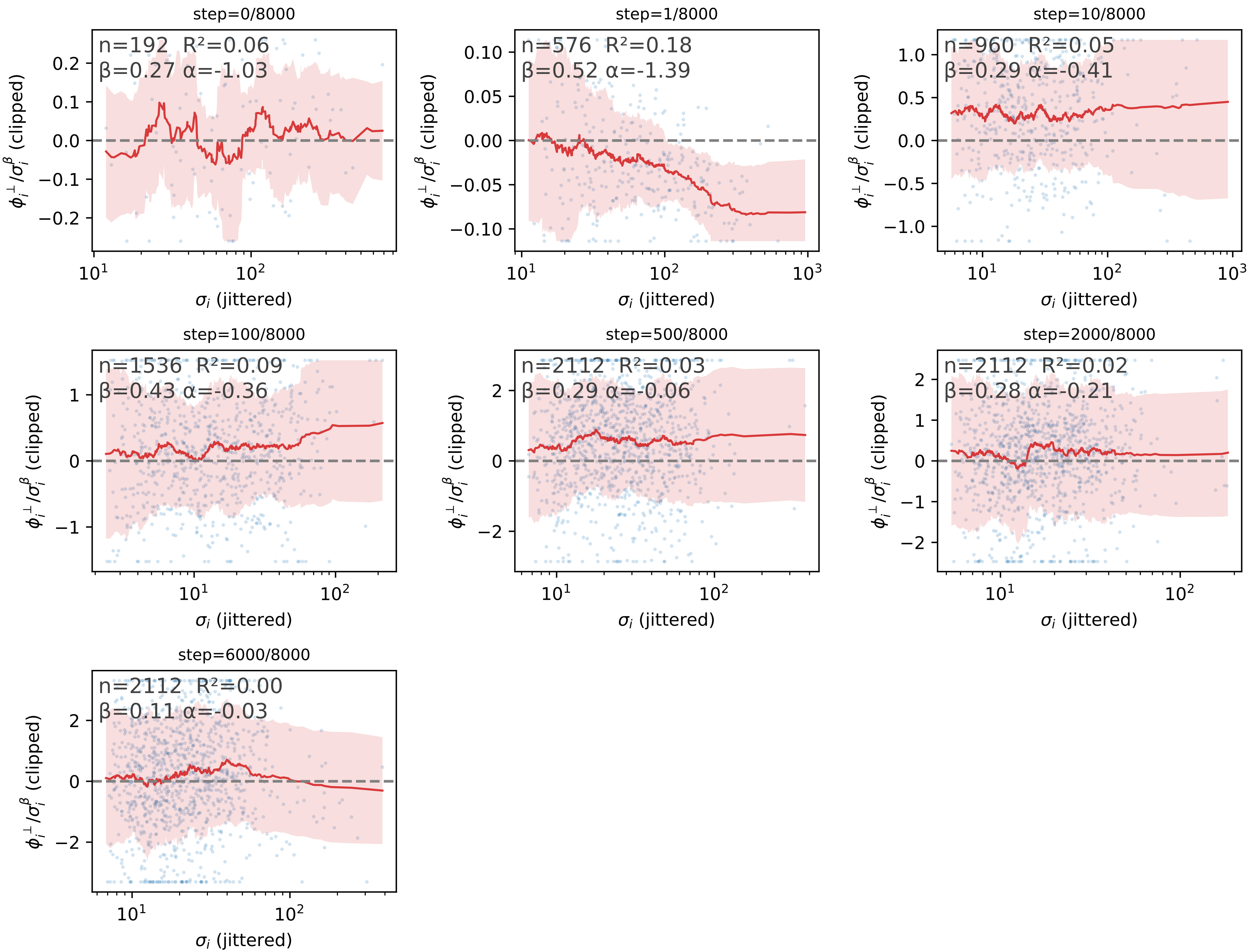
n: number of points.
R^2, β, α: least-squares fit $\log10(y) = \beta\cdot\log10(x) + \alpha$ (β the fitted power-law exponent, α the intercept).

MLP up-projection: ϕ_i/σ_i^β vs σ_i (β fit per group from $\log|\phi_i| \sim \beta \log \sigma_i$, one batch-mean point per mode per step in the group)



n: number of points.
R², β, α: least-squares fit log10(y) = β·log10(x) + α (β the fitted power-law exponent, α the intercept).

MLP up-projection: $\phi_i^\perp/\sigma_i^\beta$ vs σ_i (β fit per group from $\log|\phi_i^\perp| \sim \beta \log \sigma_i$, one batch-mean point per mode per step in the group)



n: number of points.
R², β, α: least-squares fit log10(y) = β·log10(x) + α (β the fitted power-law exponent, α the intercept).